

Assignment 3

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Preliminaries

Package Loading

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 3.2.5
```

```
library(nycflights13)
```

```
## Warning: package 'nycflights13' was built under R version 3.2.5
```

```
library(geosphere)
```

```
## Warning: package 'geosphere' was built under R version 3.2.5
```

```
## Loading required package: sp
```

```
## Warning: package 'sp' was built under R version 3.2.5
```

```
library(rgeos)
```

```
## Warning: package 'rgeos' was built under R version 3.2.5
```

```
## rgeos version: 0.3-22, (SVN revision 544)  
## GEOS runtime version: 3.4.2-CAPI-1.8.2 r3921  
## Linking to sp version: 1.2-4  
## Polygon checking: TRUE
```

```
library(haven)
```

```
## Warning: package 'haven' was built under R version 3.2.5
```

```
library(ggthemes)
```

```
## Warning: package 'ggthemes' was built under R version 3.2.5
```

```
library(maps)
```

```
## Warning: package 'maps' was built under R version 3.2.5
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 3.2.5
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:rgeos':
```

```
##
```

```
## intersect, setdiff, union
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
afg <- read.csv("~/Dropbox/courses/Mathematic Foundations of Political Methodology/Assignments/Assignment")
```

Assignment Part I

1) *Answer to Problem 1*

$$\begin{aligned} & \frac{\partial[(6x^3 + x)(x^6 - 3x^4 - 2)]}{\partial x} \\ &= (6x^3 + x) \times \frac{\partial(x^6 - 3x^4 - 2)}{\partial x} + (x^6 - 3x^4 - 2) \times \frac{\partial(6x^3 + x)}{\partial x} \\ &= (6x^3 + x) \times (6x^5 - 12x^3) + (x^6 - 3x^4 - 2)(18x^2 + 1) \end{aligned}$$

2) *Answer to Problem 2*

$$\begin{aligned} & \frac{\partial \frac{(x^2+1)}{(2x^3+1)}}{\partial x} \\ &= \frac{(2x^3 + 1) \frac{\partial(x^2+1)}{\partial x} - (x^2 + 1) \frac{\partial(2x^3+1)}{\partial x}}{(2x^3 + 1)^2} \\ &= \frac{(2x - 6x^2 - 2x^4)}{(2x^3 + 1)^2} \end{aligned}$$

3) *Answer to Problem 3*

By the composite function rule

$$\begin{aligned}\frac{d}{dx}\left(\frac{1}{v}\right) &= \frac{d}{dv}(v^{-1}) \times \frac{dv}{dx} \\ &= -\frac{1}{v^2} \frac{dv}{dx}\end{aligned}$$

By the product rule

$$\begin{aligned}\frac{d}{dx}\left(\frac{1}{v}\right) &= \frac{1}{v} \frac{d1}{dx} + 1 \frac{d}{dx} \frac{1}{v} \\ &= 0 - \frac{1}{v^2} \frac{dv}{dx} \\ &= -\frac{1}{v^2} \frac{dv}{dx}\end{aligned}$$

4) *Answer to Problem 4*

If the function is monotonic, then it is either strictly increasing or strictly decreasing.

If $f'(x) \geq 0$ for all x , and $f'(x) = 0$ for only a finite number of values of x , then f is strictly increasing.

In this case, if the function is strictly increasing, we will have $f'(x) = 5(x+a)^4 - 80 \geq 0$ for all $x \in \mathbb{R} \Rightarrow (x+a)^4 \geq 16 \Rightarrow x+a \geq 2$ or $x+a \leq -2$ for all $x \in \mathbb{R}$. However, no such a exists. We can always find a x that $-2-a \geq 2x \leq 2-a$, so that $(x+a)^4 \leq 16$.

Similarly, we can find that there is no a to make $f(x)$ strictly decreasing. Therefore, there does not exist a a to make $f(x)$ monotonic.

5) *Answer to Problem 5*

$f(x) = y = x^3 - 12x^2 + 21x + 1$, we get $f'(x) = 3x^2 - 24x + 21 = 3(x-1)(x-7)$. $f'(x) = 0$ if $x = 1$ or $x = 7$

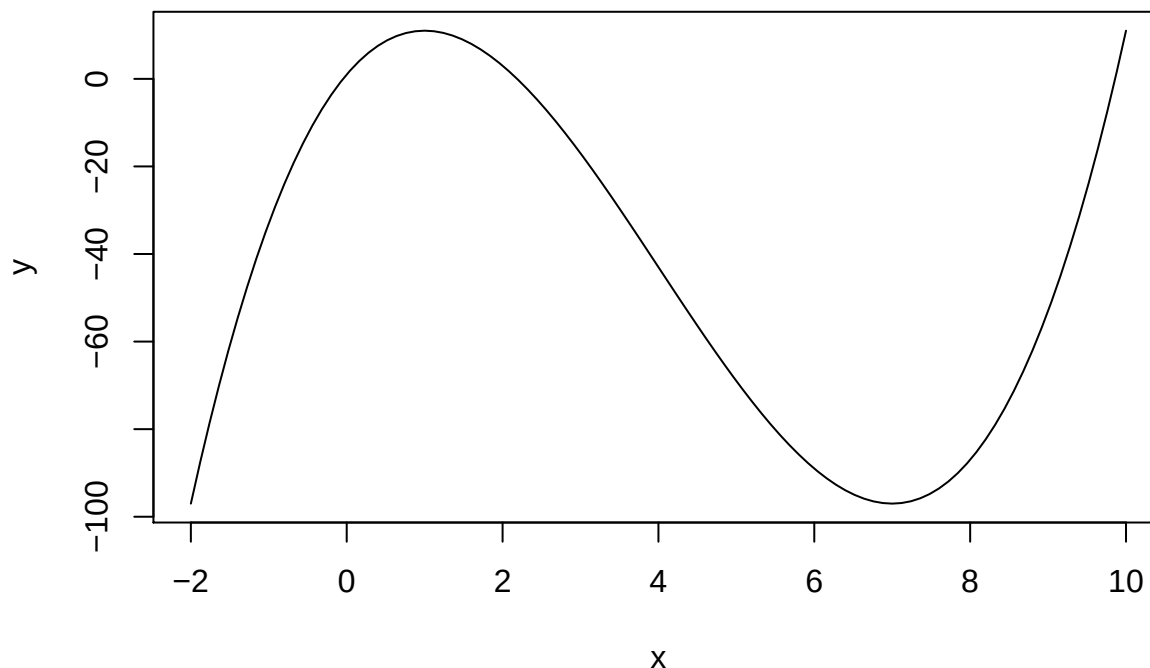
If $x = 1$, $y = 11$, If $x = 7$, $y = -97$

Thus the critical points are (1,11) or (7, -97)

Consider first the critical point (1, 11), if $x=1-$, $\frac{dy}{dx}$ is positive. If $x=1+$, $\frac{dy}{dx}$ is negative. Thus, as x goes from 1- to 1+, $\frac{dy}{dx}$ goes from + to -; (1, 11) therefore is local maximum.

Now look at the other critical point (7, -97), if $x=7-$, $\frac{dy}{dx}$ is negative. If $x=7+$, $\frac{dy}{dx}$ is also negative. As x goes from 7- to 7+, $\frac{dy}{dx}$ goes from - to +; (7, -97) therefore is a local minimum.

```
curve(x^3-12*x^2+21*x+1, from=-2, to=10, , xlab="x", ylab="y")
```



6) *Answer to Problem 6*

The points on the curve $y = f(x)$ where $f'(x) = 0$ are called critical points.

$$\frac{dy}{dx} = 5x^4 - 30x^2 + 45 = 5(x^2 - 3)^2$$

$$\frac{dy}{dx} = 0 \text{ if } x = \sqrt{3} \text{ or } x = -\sqrt{3}$$

$$\text{If } x = \sqrt{3}, y = 24\sqrt{3}, \text{ If } x = -\sqrt{3}, y = -24\sqrt{3}$$

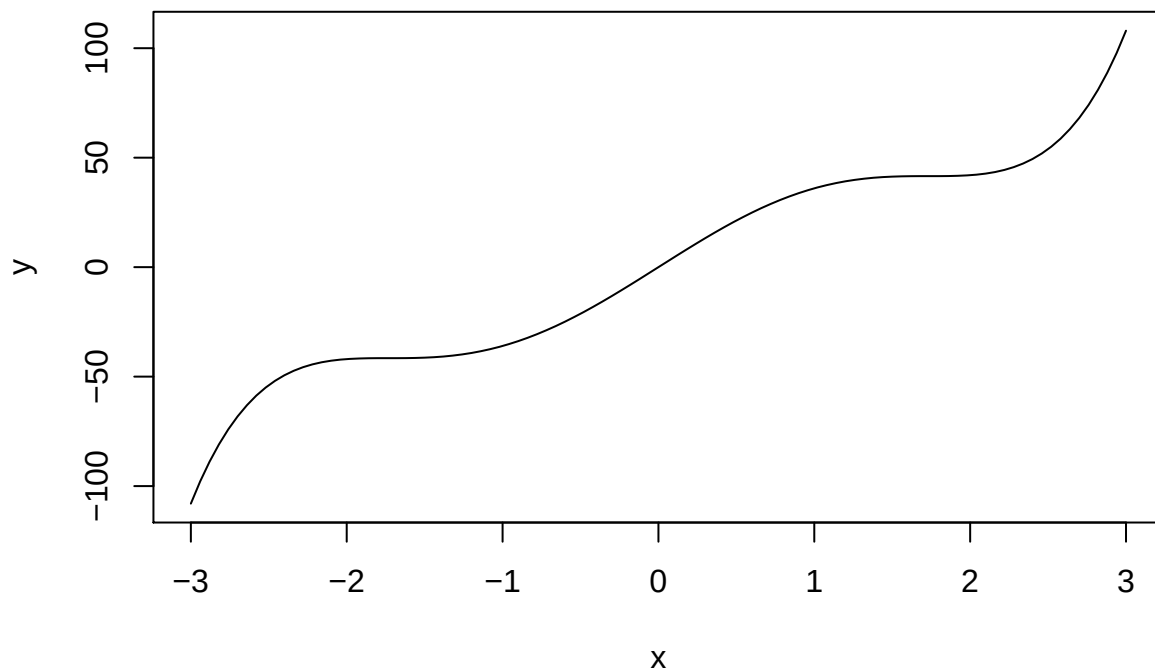
Consider first the critical point $(\sqrt{3}, 24\sqrt{3})$, if $x = \sqrt{3}-$, $\frac{dy}{dx}$ positive. If $x = \sqrt{3}+$, $\frac{dy}{dx}$ is also positive. Thus, as x goes from $\sqrt{3}-$ to $\sqrt{3}+$, $\frac{dy}{dx}$ goes from $+$ to $+$; $(\sqrt{3}, 24\sqrt{3})$ is therefore an inflection point.

Now look at the other critical point $(-\sqrt{3}, -24\sqrt{3})$, if $x = -\sqrt{3}-$, $\frac{dy}{dx}$ positive. If $x = -\sqrt{3}+$, $\frac{dy}{dx}$ is also positive. Thus, as x goes from $-\sqrt{3}-$ to $-\sqrt{3}+$, $\frac{dy}{dx}$ goes from $+$ to $+$; $(-\sqrt{3}, -24\sqrt{3})$ is therefore an inflection point.

A non-critical inflexion point is $(0,0)$.

The inverse function is $x = y^5 - 10y^3 + 45y$

```
curve(x^5-10*x^3+45*x, from=-3, to=3,xlab="x", ylab="y")
```



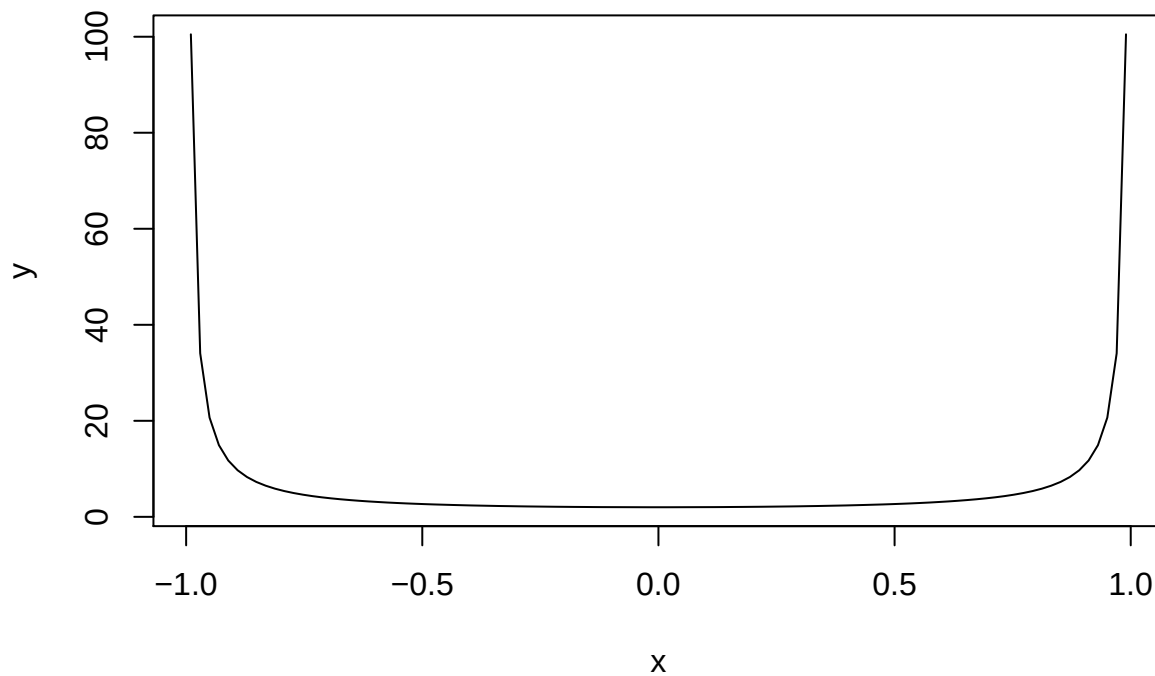
7) Answer to Problem 7

A differentiable function f is strictly convex if and only if its derivative f' is a strictly increasing function. A convex function f is strictly convex if $f''(x) > 0$ for all except a finite number of values of x .

$y = (1+x)^1 + (1+x)^{-1}$, thus $\frac{d^2y}{dx^2} = 2(1+x)^{-3} + 2(1-x)^{-3} > 0$, so the function is convex.

$\frac{dy}{dx} = \frac{4x}{(1-x^2)^2}$. For $-1 < x < 1$, $\frac{dy}{dx} = 0$ when $x = 0$. If $x = 0$, $y = 2$. The global minimum of the function is 2 at $(0, 2)$.

```
curve(2/(1-x^2), from=-0.99, to=0.99, xlab="x", ylab="y")
```



Assignment Part II

1) *Answer to Problem 1*

Plot B is clearer. If the purposes of plotting is comparing the frequency of the five categories. In the histogram, we can clearly recognized the number of Answers in five categories (if the y-axis shows the number) and compare the number of each category clearly. However, in the piechart, we can hardly compare the number of answers in each category because we can not tell which area is bigger than which, especially when the number is close.

2) *Answer to Problem 2*

a) Nuristani is spoken in the second highest altitude on average.

```
afg%>%
  group_by(LANGUAGE)%>%
  summarize(meanLati=mean(ELEVATION))%>%
  arrange(meanLati)
```

```
## # A tibble: 9 × 2
##   LANGUAGE meanLati
##   <chr>     <dbl>
## 1  Turkmen  445.7646
## 2  Balochi  534.1065
## 3   Uzbek 1022.8996
## 4  Pashto 1517.2044
## 5  Pashai 1662.4444
## 6   Other 1865.0347
## 7    Dari 2083.2627
## 8 Nuristani 2126.3806
## 9 Hazaragi 2497.7061
```

b) Balochi speakers are farthest away from Kabul on average.

```
afg%>%filter(VILLAGE_NAME_ENG == 'Kabul')%>%select(LON_X, LAT_Y)
```

```
##   LON_X  LAT_Y
## 1 69.1808 34.5228
```

```
afg%>%
  group_by(LANGUAGE)%>%
  mutate(dest = sqrt((LAT_Y - 34.5228)^2 + (LON_X - 69.1808)^2)) %>%
  summarise(distance_kabul= mean(dest))%>%
  arrange(distance_kabul)
```

```
## # A tibble: 9 × 2
##   LANGUAGE distance_kabul
##   <chr>         <dbl>
## 1  Pashai         1.088814
## 2 Nuristani     1.913830
```

```
## 3    Pashto      2.807047
## 4      Dari      3.051936
## 5 Hazaragi      3.219811
## 6      Uzbek      3.225209
## 7      Other      3.429248
## 8   Turkmen      3.736985
## 9   Balochi      7.806508
```

c) We need to calculate the distance of every village to all other villages. Find the mean of all the distance for every village and then compare.

3) *Answer to Problem 3*

a)

```
mean(c(3, 2, 1, NA),na.rm=TRUE) == 2
```

```
## [1] TRUE
```

b)

```
df <- data.frame(var1 = c(35, 40, 45))
length(df[,1])==3
```

```
## [1] TRUE
```

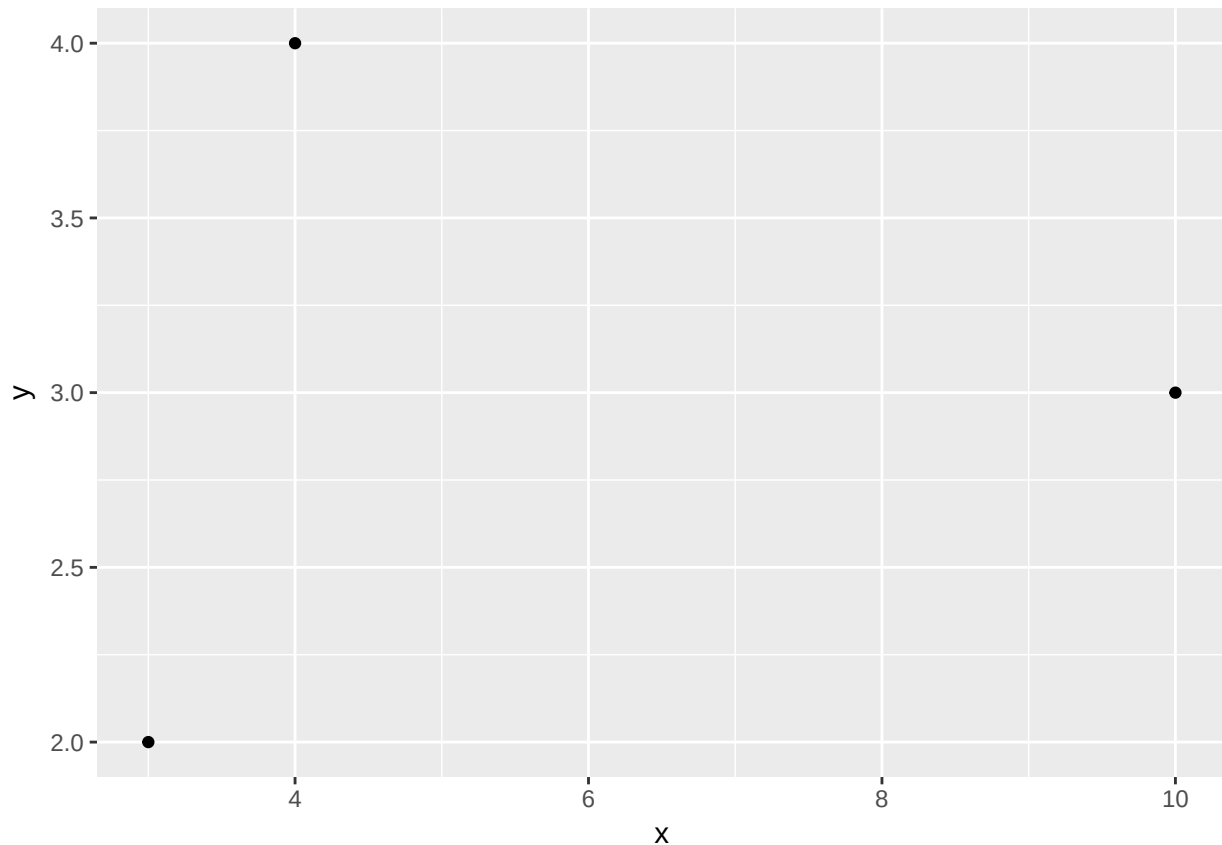
c)

```
df <- data.frame(var1 = c(35, 40, 45))
dplyr::filter(df, var1==40)==40
```

```
##      var1
## [1,] TRUE
```

d)

```
df2<-data.frame(x <- c(3, 10, 4),y <- c(2, 3, 4))
ggplot(df2,aes(x,y))+geom_point()
```



e)

```
df <- data.frame(var1 = c(35, 40, 45), var2=c("Iowa", "France", "Mexico"))
as.character(df[[1,2]])
```

```
## [1] "Iowa"
```

4) Answer to Problem 4

```
figure<-ggplot(afg, aes(LON_X, LAT_Y, color = LANGUAGE)) + geom_point(size = .05) +
  guides(color = guide_legend(override.aes = list(size = 5)))+
  labs(title="The Geography of Language Diversity in Afghanistan")+
  labs(x="Longitude", y="Latitude")+
  theme_solarized(light=FALSE)+
  theme(plot.title = element_text(hjust=0.5))+
  scale_colour_solarized("blue")
figure
```

```
## Warning: Removed 2188 rows containing missing values (geom_point).
```